Hands That Speak: Word-Level Pakistani Sign Language Translation Engine

Project Executive Summary & Technical Overview

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GitHub Repository: `https://github.com/WhatAimanWorks/Hands-That-Speak`

Executive Overview

"Hands That Speak" is an AI-powered word-level sign language translation framework engineered to bridge the communication gap between the Deaf community and the hearing world in Pakistan. The system processes video streams of isolated sign gestures, extracts spatiotemporal hand-joint and posture dynamics using modern **Vision Transformers (SignFormer, Swin Transformer, PGViT)**, and generates contextually coherent Urdu and English translation outputs via a fine-tuned **multilingual BERT (mBERT)** language model.

1. Problem Statement & Motivation

Sign language communication relies on fluid hand gestures, facial expressions, and upper-body movements. Existing assistive frameworks for Pakistani Sign Language (PSL) face critical limitations:

- **Static Gestures Only:** Most existing systems recognize isolated alphabets or static hand shapes rather than dynamic, word-level sign gestures.
- **Hardware Constraints:** Dependency on specialized depth sensors (e.g., Microsoft Kinect) prevents real-world deployment on everyday mobile smartphones.
- **Grammatical Misalignment:** Direct word-for-word gesture mapping fails to capture Urdu syntax, vocabulary variants, and contextual flow.

2. System Architecture & End-to-End Pipeline

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Raw PSL Video (30 FPS) → Pre-processing & ROI (224x224) → Keypoint Extraction
                                                                ↓
Translated Text Output ← Contextual mBERT Engine ← Spatiotemporal Vision Transformer
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A. Data Engineering & Augmentation Strategy

To overcome dataset scarcity in PSL, public benchmarks (WLPSL and Saad Butt Corpus) were augmented with volunteer-recorded isolated sign gesture sequences:

- **Spatial Augmentations:** Random cropping, 30° rotations, brightness/contrast shifts, color inversion, and greyscale conversions.
- **Temporal Augmentations:** Playback speed variations (slow/fast gesture execution).
- **Class-Wise Concatenation:** Applied iterative augmentation and retraining loops to balance class distributions and prevent majority-class bias.

B. Spatiotemporal Visual Encoders

- **SignFormer:** Employs deformable self-attention to model spatial hand configurations alongside temporal gesture transitions simultaneously.
- **Swin Transformer (Swin Tiny):** Uses shifted-window local self-attention to capture spatial details with a low computational footprint, ensuring edge-device stability.
- **Pose-Guided Vision Transformer (PGViT):** Integrates keypoint pose vectors with visual features to resolve ambiguities between visually similar hand shapes.

C. Contextual Translation Engine (mBERT)

Visual sign tokens predicted by the transformer backbones are passed into a fine-tuned Multilingual BERT (mBERT) model to convert discrete gesture tokens into syntactically correct, context-aware Urdu and English output text.

3. Empirical Results & Performance Benchmark

- **Translation Accuracy:** The proposed **SignFormer** model achieved a top **F1-Score of 0.903** and **90.28% Accuracy**, outperforming standard baseline architectures.
- **Real-Time Edge Execution: Swin Tiny** achieved inference latencies of **∼180 ms per frame**, rendering it highly stable for real-world mobile execution.
- **Environmental Noise Invariance:** Greyscale frame conversion reduced background interference, yielding an average recognition accuracy of **86%** across diverse lighting conditions.

Model Architecture	Accuracy (%)	Precision	Recall	F1-Score	Latency / Frame
SVFormer	84.50%	0.850	0.820	0.835	∼220 ms
Swin Transformer	87.20%	0.870	0.840	0.855	∼310 ms
Swin Tiny	81.78%	0.827	0.818	0.818	**∼180 ms**
SignFormer (Proposed)	**90.28%**	**0.910**	**0.903**	**0.903**	**∼200 ms**

Table 1: Empirical Model Performance Comparison

4. Mobile UI & Application Features

The system was deployed as a cross-platform mobile application built with Flutter, featuring:

1. **Real-Time Video Translation:** Supports live video capture via device camera or video file uploads for instant translation.
2. **Dual-Language Output:** Displays predicted gesture meanings in both English and Urdu (e.g., *"noisy / ΓΓΓ"*, *"alert / ΓΓΓΓΓΓ"*).
3. **Interactive Tutorial Module:** Includes a dictionary library of basic PSL tutorial clips (*Alert, Bread, Breakfast, Cheap, Chicken, College, Dangerous, Noisy, Surprise*) for practice and learning.

5. Significance for Academic & Scholarship Applications

This project highlights technical competence across:

- **Computer Vision & Deep Learning:** Implementation of state-of-the-art Spatiotemporal Vision Transformers and NLP alignment models.
- **Software Quality Assurance:** Comprehensive testing pipelines covering environment variations, background invariance, and latency optimization.

- **Social Impact:** Applied AI addressing accessibility and inclusion for underrepresented communities in developing nations.